



Solastemis™

Guardian of DNA



People make our name



About sun rays



A physical barrier between the inside and the outside, the skin is one of the most responsive organs to external aggressions, particularly to UV radiation. This latter strongly impacts the heart of skin cells, the DNA, leading to structural and functional alterations of the different skin layers.

Photoaging is the major consequence of repeated sun exposure and is associated with a particular set of visible modifications: altered microrelief, wrinkles and fine lines, coarse skin, lack of luminosity, and pigmentary disorders.

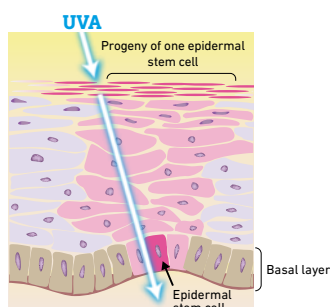
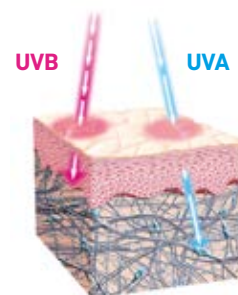
By preventing DNA and stem cells from UV-induced damage, Solastemis™ protects the cellular integrity of the skin and ensures lasting skin health and beauty.

Skin care shifts to skin health care

Ever conscious of the harsh factors impacting their skin, consumers show an affinity towards health-centric skincare products, with a specific interest in care and protection claims. In such context, the dermo-cosmetics segment is recording an increase in popularity. The future looks bright for skincare products that target lasting skin health.

Studying UVA impact on the epidermis

In the field of skin health, UV protection is a central topic. UVB was long considered to be more genotoxic than UVA. But UVA rays are now known to significantly damage both the dermis and epidermis. Their capacity to penetrate deeply in the different layers of the skin, passing the epidermis and reaching the dermis, makes them a clear enemy for the skin.



Through an in-depth research program, Gattefossé specifically uncovered that UVA can damage keratinocyte stem cells.

UVA-induced DNA lesions in keratinocytes are associated with a weakening of their self-renewal potential, called “stemness”, eventually related to functional and structural alterations in the epidermis.

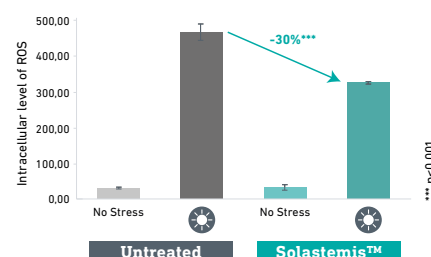
Experts in sourcing of natural products, plant chemistry and cell biology combined their knowledge to develop Solastemis™, an active ingredient able to preserve the skin from harsh UVA impact, at the epidermal and dermal level. It ensures proper homeostasis for the body's first line of defense and ultimately leads to reduce symptoms associated to photoaging.

A complete set of *in vitro* studies have been carried out in collaboration with specialists of genotoxic stress on the skin. Solastemis™ was confirmed to be a high performance active to preserve the skin from UVA radiation, fighting against oxidative stress and DNA lesions, favoring DNA repair and eventually maintaining healthy epidermal stem cells and a robust extra-cellular matrix.

Deep protection against reactive oxygen species

The generation of free radicals is the first consequence of UVA radiation. Also called ROS, they are at the origin of cascading effects and are major contributors to skin aging.

The protective activity of Solastemis™ against excessive ROS production has been assessed by measuring the level of intracellular ROS in keratinocytes after UVA exposure⁽¹⁾.



¹ Monolayer culture of proliferating normal human keratinocytes, treated with 0.5% of Solastemis™ and then irradiated with UVA (50J /cm²)

Preserving the cellular integrity of the skin

Guarding DNA from UVA radiation

Solastemis™ reduces the number of DNA lesions after UVA radiation while increasing the enzymatic repair activities of such lesions.



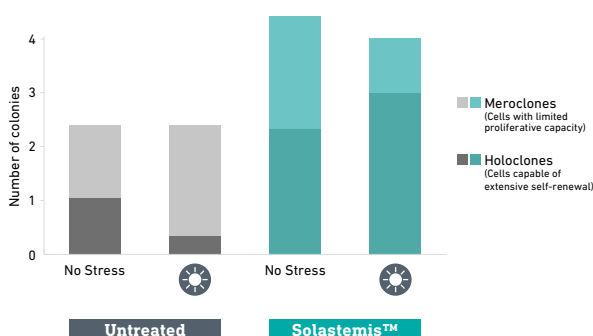
REPAIR

Enzymatic repair activities of 3 main DNA lesions ⁽³⁾



Maintenance of the epidermal stemness potential

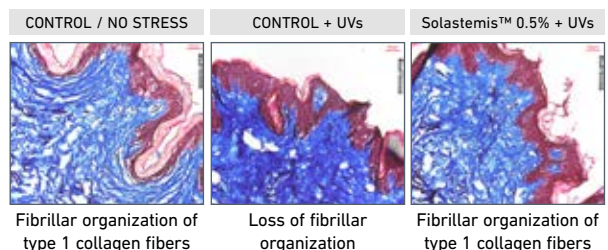
In a clonogenic assay ⁽²⁾, Solastemis™ confirmed its capacity to preserve epidermal stem cells from UVA assaults. The number of holoclones, capable of extensive self-renewal, is boosted compared to meroclones.



Dermal matrix protection

Beyond epidermis, normal human skin biopsies ⁽⁴⁾ were used to evaluate the protective efficacy of Solastemis™ on the dermal layer against the deleterious effects of solar radiation. The synthesis of extracellular matrix macromolecules and their organization are preserved.

COLLAGEN-1



+191% ****
Collagen 1 expression vs. control + UVs

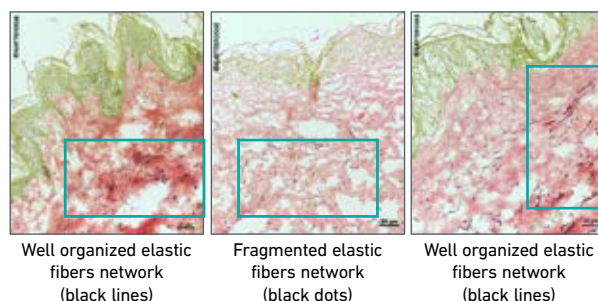
MMP-1

MMP-1 is a protease responsible for the collagen breakdown. Its excessive production following UVA exposure leads to the degradation of the extracellular matrix.

MMP-1 inhibition vs. control + UVs

-99% ****

ELASTIC FIBERS



+206% ***
Elastin expression vs. control + UVs

Solastemis™ is able to ensure a healthy skin by protecting it from UVA aggressions. In the epidermis, keratinocytes and stem cells are preserved from UVA-induced DNA damage. In the dermis, the organization and production of macromolecules are safeguarded.

⁽²⁾ Monolayer culture of proliferating normal human keratinocytes, treated with 0.5% of Solastemis™ and then irradiated with UVA (50J / cm²)

⁽³⁾ Study of enzymatic repair activity through DNA repair chips assay / 2 conditions: untreated control & Solastemis™ 0.5% (pre-treatment)

⁽⁴⁾ Skin biopsies treated for 3 days with Solastemis™ (0.5%) then irradiated with UVA (30J / cm²) and UVB (40 mJ / cm²)

**** p<0.0001 / *** p<0.001 / ** p<0.01

Photoaging-associated symptoms visibly corrected

The efficacy of Solastemis™ has been demonstrated *in vivo*⁽⁵⁾. A set of methodologies has been used to assess its capacity to reduce cutaneous symptoms associated with sun exposure and photoaging: altered microrelief and coarse skin, wrinkles and fine lines, lack of luminosity, and skin color heterogeneity.

Clear anti-wrinkle effect

CLINICAL SCORING

Crow's feet wrinkles

Variation % Dn / D0 - Solastemis™
n=25

D28

-5.8%[^]

D56

-15.2%^{*}

3D ACQUISITIONS & IMAGE ANALYSIS

Variation % D56 / D0 - Solastemis™
n=20

Area

-12.5%^{*}

Length

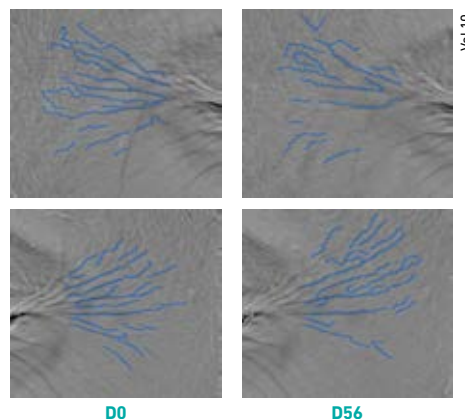
-12.5%^{*}

Volume

-12.2%^{*}

Solastemis™

Placebo



Healthy, lucent skin tone

2D ACQUISITIONS & IMAGE ANALYSIS



Variation % Dn / D0 - Solastemis™
n=20

+8.3%^{*}

ITA°

+13.3%^{*}

SELF ASSESSMENT

Variation % Dn / D0 - Solastemis™
n=25

	D28	D56
Skin radiance	18.5% [*]	29.0% [*]
Homogeneity of complexion	17.3% [*]	27.1% [*]
"Healthy glow" appearance	25.1% [^]	27.9% [*]

Tightened microrelief for reduced physiological age

CLINICAL SCORING

Variation % Dn / D0 - Solastemis™
n=25

	D28	D56
Smooth aspect	8.3%	14.4% [*]
Complexion homogeneity (texture)	9.7% [*]	12.1% [*]
Skin softness	5.6% [*]	12.8% [*]

DIAGNOSKIN®

Variation D28 / D0
n=23

Physiological age
(average)⁽⁶⁾

Solastemis™

-3.6 YEARS^{*}

Placebo

-0.8 YEARS

By offering deep protection against solar radiation, Solastemis™ efficiently corrects the signs of photoaging. Wrinkles and irregularities of the microrelief are reduced, skin color is visibly improved with a more lucent, glowing and even complexion.

⁽⁵⁾ 56-days clinical test on 33 Caucasian women, presenting photo-aged skin. Double blind study, active vs. placebo with twice daily application of the creams, on split face

⁽⁶⁾ Average physiological age (n=23): Solastemis™: 30.5 y.o. at D0 / 26.9 y.o. at D28 → -12.0%^{*} - Placebo: 31.8 y.o. at D0 / 31.0 y.o. at D28 → -2.6%

^{*} Statistical difference vs. placebo (p < 0,05) / [^] Limit statistical difference vs. placebo (0,05 ≤ p < 0,10)



Solastemis™

An active ingredient from plant origin, Solastemis™ acts at the heart of the epidermis mechanics, by protecting the DNA of keratinocytes from UVA rays and boosting endogenous DNA repair system.

Solastemis™ preserves the stem cells of the epidermis ensuring proper homeostasis for the skin's first line of defense.

It also offers a shield for the dermis by protecting key matrix proteins (collagen, elastin) from degradation. Photoaging associated symptoms are visibly corrected: wrinkles are reduced, the microrelief is smoothed, the skin complexion appears brighter, more even and healthier.

Sechium edule is an edible fruit mainly cultivated under tropical climate (also known as chayote or christophene). The fruit used to create Solastemis™ is cultivated without any chemicals and manually harvested in the majestic Cirque of Salazie, on La Réunion island. This historical supply chain shares common values aligned with Gattefossé's CSR policy: respect of biodiversity, sustainable harvesting, and continuous improvement. Solastemis™ is obtained using a NaDES composed of fructose, glycerin and water (molar ratio 1:1:5). This technology makes it possible to obtain eco-designed ingredients associated with high biological performance.

Claims

- Protects the skin from UVA-induced DNA lesions
- Preserves epidermal stem cells
- Shields the skin from photo-aging
- Smoothes the skin and visibly reduces wrinkles
- Improves skin tone homogeneity
- Provides a healthy lucent complexion

Applications

- Daily protective skin care
- Anti-aging treatments with and without SPF
- Sun care creams for face or body in addition to SPF
- Protective makeup products

INCI name: Fructose (and) Glycerin (and) Water (and) *Sechium Edule* Fruit Extract



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www.gattefosse.com



Corporate Headquarters

36 chemin de Genas - CS 70070 - 69804 Saint-Priest Cedex - **France**
+(33) 4 72 22 98 00

SOLASTEMIS™-GB-RC8-L5-22M (5949BE)

People make our name